

Rule-Based Monetary Policy in China: An Empirics-Only Counterfactual Analysis

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Abstract

Since 2023, Chinese CPI inflation has hovered near zero, persistently undershooting the targets set in the annual Government Work Report. This paper applies the empirics-only counterfactual framework of Caravello, McKay and Wolf (2025) to ask what would have happened to Chinese inflation, output, and the exchange rate if the People’s Bank of China had instead followed an explicit targeting rule over this period. Two sets of impulse response functions to monetary policy shocks — one identified from a narrative policy-rule residual on the 7-day repo rate (FR007), the other from high-frequency FR007 surprises on policy event dates — are combined with a 7-variable Bayesian VAR baseline projection in a stacked least-squares solver to recover the shock sequence that best implements alternative targeting rules. The headline finding is a persistent gap between counterfactual paths and the Government Work Report CPI targets, even under aggressive strict targeting, suggesting that conventional interest rate policy alone could not have closed the realized inflation gap. The result is robust to flexible targeting that includes output and exchange-rate stabilization objectives, and to extending the estimation sample to 2025. A secondary finding — a positive short-run CPI response to a contractionary FR007 shock that appears in both identification schemes — is interpreted as a substantive feature of China’s CPI methodology, in which floating-rate mortgage interest enters the housing sub-index, rather than as a model deficiency.

Keywords: monetary policy, counterfactuals, China, inflation targeting, sufficient statistics.

JEL codes: E52, E58, E31.

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1 Introduction

Since 2023, Chinese CPI inflation has hovered near zero, persistently undershooting the official targets set in the annual Government Work Report (GWR). The GWR CPI targets for 2023, 2024, and 2025 were 3%, 3%, and roughly 2%, respectively; realized year-on-year CPI growth over the same period averaged close to zero. Producer prices have remained in negative territory since late 2022, and the deflationary stance has coincided with severe liquidity stress in the real estate sector — exemplified by the bankruptcies and near-defaults of major developers such as Evergrande and Country Garden — and with elevated local government financing-vehicle (LGFV) debt risks.

Despite these pressures, the People’s Bank of China (PBoC) has maintained a cautious, gradualist stance. The Loan Prime Rate (LPR), now the principal anchor for new loan pricing, has been lowered only moderately, from 3.65% in 2022 to roughly 3.0% in early 2026, a pace that has been widely viewed as insufficient to arrest the deflationary spiral.² The 2024 Central Economic Work Conference (CEWC) introduced for the first time the formulation of “a reasonable rebound in the price level” (*wujia heli huisheng*, 物价合理回升); the 2025 CEWC explicitly tied this objective to the conduct of monetary policy. These developments raise a fundamental question about the recent past: was the discretionary failure to meet the inflation target a result of structural constraints on monetary policy in China, or was it primarily a policy choice? Equivalently, what would the trajectory of Chinese CPI, output, and the exchange rate have looked like over 2023–2025 if the PBoC had instead followed an explicit, rule-based targeting policy?

Approach. This paper answers that question by applying the empirics-only counterfactual framework of Caravello et al. (2024) to China.³ The approach builds on the sufficient-statistics insight of McKay and Wolf (2023) and Barnichon and Mesters (2023): under weak structural assumptions, evaluating macroeconomic outcomes under alternative monetary policy rules requires only two empirically estimable objects — a reduced-form baseline projection of the economy and the causal impulse-response functions (IRFs) of macro variables to monetary policy shocks. CMW2025 show that, when the contemplated counterfactual involves changes to the short end of the yield curve, these objects can be taken directly from the data, sidestepping the need for a fully specified structural model. The framework is therefore well-suited to settings where the underlying structure is poorly known — arguably the case for China, where multi-

²See, for example, the discussion of monetary policy options in the IMF’s Article IV consultations with China for 2024 and 2025.

³Throughout the paper I refer to (Caravello et al. 2024) as “CMW2025” and to the foundational result in (McKay and Wolf 2023) as “MW2023”.

instrument policy, dual-track interest rates, and tight monetary–fiscal coordination complicate any DSGE-style identification of mechanisms.

The empirical implementation proceeds in four blocks, in the spirit of the CMW2025 application to U.S. post-COVID inflation. First, a 7-variable Bayesian VAR with Minnesota prior, estimated on monthly data from 2002 onward, produces a baseline projection of the Chinese economy from a chosen forecast date. Second, narrative monetary policy shocks are identified as the residual of a forward-looking policy rule for FR007 that combines elements from Chen et al. (2025) (asymmetric output gap, lagged interest rate) and Miranda-Agrippino et al. (2026) (post-2006 exchange-rate gap term). Third, high-frequency shocks are constructed as daily close-to-close FR007 changes on dates of PBoC announcements that involve a change in one of the core policy instruments — 7-day reverse repo, 1-year MLF, 1-year LPR, RRR for large financial institutions, and pre-2015 benchmark deposit/lending rates — and aggregated to monthly frequency. Fourth, the shock sequence that minimizes a weighted quadratic loss over CPI deviations from the GWR target, output deviations from the GWR growth target, FR007 first differences (a smoothness penalty), and policy-induced deviations of the nominal effective exchange rate from baseline is recovered from a stacked-OLS problem in which both narrative and HFI transmission maps enter as columns of the policy-shock causal-effect matrix.

Findings. Three results emerge.

First, even under strict CPI targeting — in which the loss function places weight only on the inflation gap and an interest-rate smoothness penalty — the counterfactual CPI path remains substantially below the GWR targets throughout 2023–2025. Adding flexible targeting motives (output stabilization, exchange-rate stability) tightens the gap further. The relevant comparison here is between counterfactual CPI and the GWR target, not between counterfactual and baseline: the BVAR baseline is itself notably pessimistic from 2023 onward, likely because the model extrapolates post-COVID deflationary dynamics that may represent a structural break from the estimation sample. The persistent gap between the counterfactual and the announced target is the central finding.

Second, the counterfactual reveals a clear policy trade-off. Across all targeting rules considered, more aggressive rate cuts achieve lower (more negative) counterfactual inflation deviations, consistent with a price-puzzle pattern in the underlying IRFs (see below); the cost is larger NEER swings and greater output volatility relative to more balanced rules.

Third, both identification schemes deliver an initial *positive* CPI response to a contractionary FR007 shock — a price puzzle. Rather than treating this as a deficiency

to be patched away (e.g., by adding commodity-price controls or sign restrictions), I follow Chodorow-Reich (2025) in interpreting it as a substantive feature of Chinese CPI methodology. The Chinese CPI’s housing sub-index (居住类) employs a user-cost approach that includes mortgage interest payments; with the bulk of Chinese mortgages on floating rates pegged to the LPR, an upward FR007 shock that propagates to the LPR mechanically raises the housing sub-index in the short run. The puzzle is therefore informative about the housing channel of monetary transmission, not a sign that the shocks are misidentified.

Contributions. This paper makes three contributions.

First, it is, to my knowledge, the first application of the empirics-only counterfactual framework of Caravello et al. (2024) to a non-U.S., emerging-market context.⁴ The application is non-trivial: the framework was originally designed around the federal funds rate as a single, well-identified policy instrument, while the PBoC operates a multi-instrument framework with dual-track rates. The paper takes a clear stand on this issue by selecting FR007 — the market rate that anchors HFI surprises and that Chen et al. (2025) use as the dependent variable in their policy rule — as the unifying instrument across all four building blocks.

Second, the paper combines two distinct shock-identification schemes within a single counterfactual exercise. Following the CMW2025 logic, both narrative and HFI shocks enter as columns of the transmission map, allowing the solver to choose a mixture rather than treating one as primary and the other as robustness. This is more demanding than the standard practice in the China monetary-policy literature, where a single shock series is typically estimated and compared across alternatives.

Third, the paper documents and rationalizes a robust price puzzle in Chinese CPI responses to monetary policy shocks. The interpretation through the housing user-cost channel, building on Chodorow-Reich (2025), has implications for the design of future inflation-targeting frameworks in China and for the cross-country literature on monetary transmission to housing.

Roadmap. Section 2 reviews the related literature on monetary policy identification in China, on counterfactual policy evaluation methods, and on the recent Chinese deflationary episode. Section 2 provides institutional background on the PBoC’s policy framework and on the Government Work Report target system. Section 4 sets out the empirical framework, including the BVAR baseline projection, the two shock-identification schemes, and the counterfactual solver. Section 5 describes the data. Section 8.1

⁴Bouscasse and Hu (2024) apply the related sufficient-statistics framework of (McKay and Wolf 2023) to fiscal counterfactuals in the U.S. post-COVID inflation episode; the present paper is closer in spirit to that exercise than to the original U.S. monetary applications in CMW2025.

presents the estimated impulse responses and discusses the price puzzle. Section 8.3 reports the main counterfactual results. Section 9 discusses limitations and avenues for future research. Section 10 concludes.

2 Chinese Context

2.1 PBoC: From Subordinate Agency to Independent Institution

Established in 1948, the PBoC operated for its first three decades as a functional arm of the Ministry of Finance (MoF) rather than as an independent central bank. It was not until 1983 that the State Council formally recognized PBoC's central bank status, and only through the Law of the People's Bank of China passed by the National People's Congress in 1995 that legal separation from MoF was achieved (Zheng and Wang, 2021). Prior to 1995, PBoC's budget was entirely controlled by MoF, which retained financial and, to a degree, personnel authority over its operations; after independence, PBoC was elevated to ministerial rank under the State Council and gradually acquired full budgetary autonomy by the mid-2000s. The 1995 reform also prohibited PBoC from purchasing treasury bonds in the primary market, severing MoF's ability to fund fiscal deficits directly through the central bank. This legislative change was not the product of disinterested institutional design: the dominance of MoF and the consequent over-reliance on fiscal policy had contributed to the overinflation and economic crises of the late 1980s and early 1990s, and the independence reform reflected a deliberate recalibration of power within the central government toward technocratic control of monetary conditions (Zheng and Wang, 2021; Bell and Feng, 2014).

A parallel organizational reform in 1998 insulated PBoC from local government pressure by replacing the system of locally embedded branches with nine regional branches modeled on the Federal Reserve, each directly reporting to PBoC headquarters in Beijing. This removed the formal channels through which local civil and party authorities had routinely intervened in local financial institutions, contributing to the excessive investment and overheating of the mid-1990s. More recently, however, the regional branch system has itself been phased out: as provincial-level PBoC branches have become sufficiently independent of local government influence, and as the regional branches proved unable to develop the research and policy formulation capabilities originally envisioned, the State Council moved toward a system of direct provincial reporting to Beijing, rendering the intermediate regional tier redundant.

A further wave of functional narrowing — the separation of supervisory responsibilities into the China Banking Regulatory Commission, China Securities Regulatory

Commission, and China Insurance Regulatory Commission in 2003, later consolidated in 2018 — allowed PBoC to concentrate on monetary policy and financial stability. Despite these accretions of formal autonomy, important constraints remain: monetary policy authority ultimately resides with the State Council, and the Monetary Policy Committee retains only an advisory function. As Miranda-Agrippino, Nenova, and Rey (2026) note, substantive policy decisions must be approved by the State Council, which structurally limits the timeliness and forward-looking character of PBoC’s policy communication. This institutional setting — formal independence combined with constrained discretion — is the backdrop against which the counterfactual exercise in this paper should be interpreted: the counterfactual paths quantify the economic implications of a shift toward rule-based inflation targeting, not a description of what PBoC could have achieved unilaterally within the existing political structure.

2.2 Evolution of Monetary Policy Tools

The transformation of PBoC’s policy toolkit over the past three decades reflects both its institutional evolution and a set of structural features specific to China’s financial system. In the earlier part of the sample, PBoC relied primarily on quantity-based instruments — chiefly the Reserve Requirement Ratio (RRR), administratively set benchmark lending and deposit rates, and direct guidance over M2 and total credit growth. The preference for these tools was not simply inertia or imitation of an older Fed playbook; it reflected practical constraints on price-based transmission. China’s interest rate markets were heavily repressed for much of this period, with binding ceilings on deposit rates and floors on lending rates that limited the scope for market interest rates to serve as effective policy signals. In an environment where bank lending was allocated partly by administrative guidance and partly through state-owned banks with soft budget constraints, controlling the quantity of credit was a more reliable lever than adjusting its price (Das and Song, 2022). PBoC’s mandate compounded this: unlike central banks with a single inflation target, PBoC operated under multiple official objectives — price stability, economic growth, employment, exchange rate stability, and financial market development — which favored a toolkit broad enough to address each dimension separately (Miranda-Agrippino, Nenova, and Rey, 2026). M2 growth served as the primary intermediate target through the mid-2010s, with the RRR functioning as a blunt but effective instrument for managing aggregate bank liquidity. From 2012, as the rise of shadow banking eroded the correspondence between M2 and broader credit conditions, PBoC shifted its intermediate target toward Total Social Financing (TSF), which better captured off-balance-sheet and non-bank lending (Chen, Ren, and Zha, 2018). During the WTO-era surplus years, Central Bank Bills were added as a sterilization instrument to absorb the excess liquidity generated by foreign exchange

inflows, and were phased out after 2013 as the exchange rate regime was liberalized and the sterilization motive diminished.

The transition toward price-based policy accelerated from 2013 onward, driven by the broader financial liberalization agenda of the 13th Five-Year Plan and a recognition by PBoC itself that the correlation between M2 and real activity was weakening as the economy became more market-oriented (Yi Gang, 2018). PBoC introduced the Standing Lending Facility (SLF) and the Medium-Term Lending Facility (MLF) to manage short- and medium-term liquidity respectively, with the SLF rate forming the corridor ceiling. The completion of deposit rate liberalization in October 2015 and the progressive elevation of the 7-day reverse repo rate (FR007) as the operational short-term anchor marked the emergence of a two-tier interest rate corridor. The reform of the Loan Prime Rate (LPR) in August 2019, reconstituting it as a spread over the 1-year MLF rate, extended price-based transmission into the retail credit market. By the latter part of the sample, the composite monetary policy indicator (CMPI) of Miranda-Agrippino, Nenova, and Rey (2026) co-moves primarily with FR007, the 3-month SHIBOR, and the LPR, while RRR adjustments continue as a supplementary structural liquidity tool. Chen, Xiao, and Zha (2025) and Das and Song (2022) both build their shock identification around FR007 movements, reflecting its status as the primary observable policy rate in the post-2015 framework.

Despite this progress, several institutional and structural features continue to constrain monetary policy transmission. Most fundamentally, PBoC lacks full instrument independence: under Article 5 of the PBC Law, decisions on money supply, interest rates, and exchange rates must be approved by the State Council before implementation. Monetary policy decisions are therefore often taken collectively as part of broader policy packages rather than on the basis of PBoC's independent cyclical assessment. Das and Song (2022) show this has direct macroeconomic consequences: the impact of monetary policy shocks that are *not* accompanied by concurrent fiscal measures is significantly weaker and less persistent than that of coordinated ones, implying that much of the apparent effectiveness of Chinese monetary policy in the data reflects fiscal amplification rather than the independent transmission of the interest rate channel. The Monetary Policy Committee compounds this constraint: it functions purely as a consultative body, meets quarterly without pre-announced dates, and cannot credibly commit to forward guidance, leaving the price-based framework without the expectations management tools that are central to monetary transmission in advanced economies.

A further development, increasingly prominent after 2015 and particularly since 2020, is the proliferation of so-called structural monetary policy tools (结构性货币政策

策工具). Instruments such as the Pledged Supplementary Lending (PSL), re-lending (再贷款), and re-discounting (再贴现) facilities direct subsidized central bank liquidity at designated sectors — agriculture, small and micro enterprises, green investment, shantytown renovation, and most recently real estate support — at preferential rates set below the standard MLF rate. Unlike conventional open market operations, which adjust the aggregate price of credit, these instruments create a dual-tier credit market in which targeted borrowers receive funding whose cost is administratively determined. This represents a progressive blurring of the boundary between monetary and fiscal policy: PBoC is effectively allocating credit across sectors according to government industrial and social priorities, with the implicit fiscal subsidy absorbed through below-market lending rates rather than explicit budget outlays. The growing scale of these facilities raises a non-trivial identification challenge for the latter part of the sample: aggregate PBoC balance sheet expansion increasingly conflates conventional monetary easing with directed industrial policy financing, complicating any attempt to characterize the cyclical stance of monetary policy from observable instrument movements alone.

2.3 Recent Policy Background

2.3.1 The Government Work Report Target System

China’s CPI has hovered around 0% since 2023, against a government target that stood at “around 3%” for most of the preceding decade — a gap that is the immediate motivation for this paper’s counterfactual exercise. To interpret this gap correctly requires understanding the institutional status of that target. The Government Work Report (GWR), delivered by the Premier to the National People’s Congress each March, sets annual numerical objectives for GDP growth, CPI inflation, employment, and the fiscal deficit. These are not legally binding commitments: the PBoC’s mandate under the 2003 Law on the People’s Bank of China is formulated as “maintaining the stability of the value of the currency and thereby promoting economic growth,” with no quantitative CPI target embedded in statute. The GWR targets thus function as an implicit accountability device — a public signal of the central government’s priorities that shapes expectations and provides a reference point for policy evaluation — rather than a formal inflation-targeting mandate of the kind analyzed in the Bernanke-Reinhart tradition. This distinction matters for how the counterfactual exercise in this paper is framed: we are not asking what would have happened under a New Zealand-style legally embedded inflation target, but what would have happened had PBoC treated the GWR’s own stated CPI objective as a binding operational constraint. Alongside the GWR, the Central Economic Work Conference (CEWC), convened each December and chaired by the General Secretary rather than the Premier, carries at least equivalent

political authority as the highest-level annual economic policy document under the CPC leadership. The CEWC sets the agenda that the subsequent GWR then operationalizes; its signals on monetary and price objectives therefore serve as an advance indicator of policy priorities. The 2024 CEWC notably called for promoting a rebound in the price level to a reasonable range, and the 2025 CEWC went further by directly linking monetary policy easing to this objective — the first time the General Secretary had connected the rate instrument to a price-level goal in this explicit manner. The 2025 GWR accordingly lowered the CPI target from “around 3%” to “around 2%”, retained at “around 2%” in the 2026 GWR. This revision is itself interpretively ambiguous: it can be read as a concession to the structural disinflationary environment and an implicit acknowledgment that 3% was unrealistic, or — more charitably — as a more achievable and thus operationally more serious commitment to a target that the government intends to pursue rather than merely announce. The paper’s loss function takes the latter interpretation as its baseline.

One additional justification for departing from the conventional Taylor rule framework — which measures deviations from unobservable potential output and the natural rate of interest — is that the GWR’s explicit numerical objectives for both GDP growth and CPI inflation make it possible to construct the policy loss function entirely from observable, publicly announced targets. This is not merely a pragmatic convenience: Beckworth and Hendrickson (2020) show in a New Keynesian model with imperfect central bank information that output gap forecast errors by the Federal Reserve account for up to 13% of output gap volatility, and that nominal income targeting dominates the Taylor rule precisely because it avoids this source of policy error. Garín, Lester, and Sims (forthcoming) confirm that nominal GDP targeting outperforms both inflation targeting and a conventional Taylor rule in welfare terms, particularly when supply shocks are present and wages are sticky relative to prices — conditions broadly applicable to China’s post-pandemic disinflationary episode. The GWR’s combined growth and inflation targets are structurally equivalent to a nominal income target, and the present paper’s use of those targets as the loss function benchmark can be understood as operationalizing this insight in the Chinese institutional context.

2.3.2 The 2023-2025 Deflationary Episode in Context

China’s producer price index turned negative in October 2022 and remained so through the end of the paper’s sample, while consumer price inflation fell to near zero by early 2023, briefly turned negative in mid-2023, and hovered around 0% in both 2023 and 2024. The proximate drivers are well-documented: the post-COVID demand shortfall, the contraction of the property sector following the Evergrande default in late 2021 and the subsequent distress of Country Garden and local government financing vehicles, and

the associated collapse in household wealth and consumer confidence. PBoC’s policy response was cautious: the 1-year LPR was trimmed from 3.65% to 3.0% over the period, RRR was cut twice, and structural tools were expanded, but no episode of aggressive easing comparable to prior cycles was undertaken. This restraint is notable against the backdrop of PBoC’s own history. In 2008–09, the central bank combined deep benchmark rate cuts with successive RRR reductions, providing monetary support for the RMB 4 trillion fiscal stimulus. In 2015–16, the “double-cut” cycle — simultaneous reductions in benchmark rates and RRR deployed six times over eighteen months — was the response to the stock market crash and growth slowdown. Even in 2020, though the COVID response was more measured than the Federal Reserve’s, PBoC enacted targeted RRR cuts, MLF rate reductions, and a significant expansion of structural re-lending facilities within weeks of the initial lockdowns. The 2023–25 episode, by contrast, is characterized by a slow, incremental loosening that remained insufficient to close the gap against official CPI targets throughout the period. This is not, therefore, an episode in which the counterfactual question is vacuous — the observed policy path was not obviously the constrained optimum, and the institutional and political constraints documented in Section 2 provide a plausible account of why a more aggressive response was not forthcoming.

2.3.3 Connecting to the Counterfactual Literature

The institutional and operational history reviewed above establishes the empirical context for this paper’s central exercise. Having characterized what PBoC did, through what instruments, and against what targets, the following section reviews the literature on macroeconomic policy counterfactual analysis that provides the methodological framework for asking what it should have done instead.

3 Literature Review on Macroeconomic Policy Counterfactuals

Evaluating the macroeconomic effects of monetary policy faces two interlocking problems. First, policy responds endogenously to the state of the economy, so the observed correlation between policy variables and macroeconomic outcomes cannot be interpreted causally without identifying restrictions. Second, even if the causal effects of unanticipated policy actions can be identified, the relationships between policy and outcomes are not generally invariant to changes in the systematic component of policy — this is the Lucas (1976) critique. A counterfactual exercise is meaningful only insofar as the relationships used to construct it remain valid under the contemplated change in the policy rule.

Two parallel literatures have responded to these problems. The first, organized around vector autoregressions (VARs), seeks to identify policy shocks while remaining as agnostic as possible about deep structural parameters. The second, organized around the New Keynesian (NK) research program, builds fully specified microfounded models in which the structural parameters are by construction invariant to the policy rule. A recent third line of work bridges these approaches by establishing sufficient-statistics conditions under which counterfactuals can be obtained from reduced-form objects alone. This paper sits within the third tradition and applies its empirics-only variant to China.

3.1 The reduced-form VAR tradition

The modern empirical literature on monetary policy traces to Sims (1980), who proposed VARs as an alternative to the large-scale structural macroeconometric models that had dominated the field, on the grounds that their identifying restrictions were “incredible.” Within this tradition, a substantial sub-literature has developed methods to identify exogenous variation in policy. Two principal approaches have emerged. Narrative methods, originating in Romer and Romer (2004), residualize policy decisions against the systematic policy rule by exploiting documentary evidence on policy intentions; high-frequency identification (HFI) methods, originating in Kuttner (2001) and extended in Gürkaynak, Sack and Swanson (2005) and Nakamura and Steinsson (2018), exploit movements in market rates within tight windows around policy announcements. The identified shock series are typically used as external instruments in SVAR settings (Mertens and Ravn, 2013; Stock and Watson, 2018) to recover impulse response functions.

Both shock-identification approaches face ongoing challenges. The most prominent is the information-effect critique: that even properly identified policy shocks may convey private central-bank information about future economic conditions, so the responses of macro variables to “shocks” reflect both the direct policy impulse and an information channel (Nakamura and Steinsson, 2018; Jarociński and Karadi, 2020). Recent work pushes back on this interpretation, arguing that what appears as an information effect is largely a response to predictable economic news (Miranda-Agrippino and Ricco, 2021; Bauer and Swanson, 2023). Because both methods are central to the methodology of this paper, Section 4 returns to them in greater detail and surveys the literature on their application to China; the present discussion focuses on counterfactual evaluation.

Once policy shocks are identified, the natural next step in the VAR tradition is to construct counterfactual paths by manipulating the shock sequence in an estimated SVAR. Bernanke, Gertler and Watson (1997) implement this approach for oil-price shocks, asking how much of the post-shock decline in U.S. output is attributable to

the systematic monetary response rather than to oil prices directly. Sims and Zha (1995, 2006) develop the canonical multi-variable identified VAR and use it to evaluate alternative U.S. policy regimes. The principal difficulty with this strategy is that VAR-based counterfactuals constructed by repeatedly drawing policy shocks to implement an alternative rule remain vulnerable to the Lucas critique. If private agents had expected the alternative rule to prevail, the reduced-form coefficients of the VAR — including the relationship between shocks and outcomes — would generally differ. Benati (2010) and Beraja (2024) document that this problem is non-negligible in calibrated NK environments. Reduced-form counterfactuals are thus best understood as informative about modest deviations from the prevailing rule, rather than as evaluations of large or persistent regime changes.

3.2 The structural NK tradition

The parallel response to the Lucas critique builds models in which agents optimize given the policy rule, so that the parameters describing preferences and frictions are by construction invariant to policy. Christiano, Eichenbaum and Evans (2005) and Smets and Wouters (2007) established the medium-scale New Keynesian DSGE framework — with nominal rigidities, habit formation, and variable capital utilization — as the workhorse for monetary policy counterfactual analysis in academia and central banks. Kaplan, Moll and Violante (2018) extend this framework to heterogeneous agents (HANK) and show that the transmission mechanism qualitatively changes when household balance sheets and incomplete markets are explicitly modeled: indirect general-equilibrium effects on labor income dominate the direct intertemporal substitution channel that drives transmission in representative-agent settings. Auclert, Bardóczy, Rognlie and Straub (2021) develop sequence-space Jacobian methods that make HANK estimation computationally tractable, and a growing literature applies the framework to substantive policy questions.

The structural approach overcomes the Lucas-critique vulnerability, but at a substantial cost: counterfactual conclusions depend on which structural model is chosen, and the relevant choices — the form of household heterogeneity, the degree of market incompleteness, the nature of nominal and real frictions — are difficult to discipline from the aggregate data alone. Different specifications can deliver materially different counterfactual answers to the same policy question.

3.3 Sufficient statistics and the empirics-only revival

A recent line of work bridges these two traditions. McKay and Wolf (2023) and Barnichon and Mesters (2023) establish that, under invertibility, the impulse responses of macroeconomic variables to a set of contemporaneous and anticipated (“news”) shocks

to the prevailing policy rule — together with a baseline projection of the economy — constitute a sufficient statistic for counterfactual policy evaluation under a wide class of alternative rules. The key idea is that the counterfactual is constructed not by changing the policy rule and re-propagating through the estimated reduced-form dynamics — the strategy of Sims and Zha (1995, 2006), which encounters the Lucas critique because the reduced-form dynamics themselves depend on the prevailing rule — but by implementing the alternative policy path through a sequence of policy and news shocks, all occurring within the prevailing rule. Under linearity, this within-regime construction is mathematically equivalent to the across-regime counterfactual to first order, and the empirically estimated IRFs already encode the correct private-sector response. The validity of this equivalence depends on the contemplated counterfactual remaining within the span of historically observed policy movements; the scope of this assumption is discussed in Section 4. Caravello, McKay and Wolf (2025) extend this insight by showing that, when the contemplated counterfactual involves changes only at the short end of the yield curve, the required IRFs can be estimated purely empirically, sidestepping the need to commit to a particular structural model. Bouscasse and Hong (2024) apply a closely related sufficient-statistics framework to evaluate counterfactual fiscal rules in the United States.

The sufficient-statistics literature thus offers an alternative to both horns of the structural-vs-reduced-form trade-off. It retains the empirical discipline of the VAR tradition while overcoming the Lucas-critique vulnerability of standard VAR-based counterfactuals, and it delivers counterfactual paths comparable to those of fully specified NK models without committing to one. The empirical demands are however substantial: the framework requires reliable estimates of the causal effects of both contemporaneous and anticipated policy shocks, which is a high bar in settings where policy is conducted through multiple instruments or where private-sector expectations are difficult to measure.

3.4 Position of this paper

This paper applies the empirics-only framework of Caravello, McKay and Wolf (2025) to evaluate counterfactual Chinese monetary policy over the 2023-2025 deflationary episode. The exercise serves two purposes. First, it stress-tests the framework outside its original U.S. context, in a setting where policy is conducted through a multi-instrument operating framework, where fiscal-monetary coordination is institutionalized, and where the structural transmission of policy is incompletely understood — conditions that arguably make the empirics-only approach more attractive than the structural alternative, but that also raise concerns about whether its identification requirements can be met. Second, it produces a substantive answer to the question of whether a

rule-based PBoC could have closed the realized inflation gap relative to the announced Government Work Report targets, or whether the realized inflation undershoot reflects deeper structural constraints on monetary policy in China.

The identification of Chinese monetary policy shocks, which feed the framework, draws on a parallel literature that has adapted both narrative and high-frequency methods to the Chinese institutional setting. That literature — covering work by Chen, Ren and Zha (2018); Chen, Xiao and Zha (2025); Miranda-Agrippino, Nenova and Rey (2026); Das and Song (2022); He, Jia, Li and Wu (2024); and Gürkaynak, İpek and Ricco (2026), among others — is reviewed in Section 4 alongside the implementation details of the methodology.

4 Empirical Strategy

4.1 The CMW Sufficient-Statistics Approach

The standard approach to evaluating monetary policy counterfactuals since Lucas (1976) has been fully structural: specify a model, change the policy rule, and re-solve. The sufficient-statistics approach of McKay and Wolf (2023), Barnichon and Mesters (2023), and Caravello, McKay and Wolf (2025, henceforth CMW) offers an alternative that sidesteps the need for a complete structural model while remaining robust to the Lucas critique.

The key distinction from earlier VAR-based counterfactual exercises — such as Sims and Zha (1995, 2006) and Bernanke, Gertler and Watson (1997) — is in how the counterfactual policy is implemented. The earlier literature *adjusted the VAR coefficients* to reflect an alternative policy rule, which is vulnerable to the Lucas critique: agents' behavior and expectations would change under the new rule, invalidating the original reduced-form coefficients. CMW instead *hold the estimated impulse response functions fixed* and treat the policy rule change as a linear combination of monetary policy shocks at each period. The counterfactual path is then simply the baseline forecast plus the transmission map applied to a shock sequence that minimizes a loss function encoding the alternative rule. Because the shocks are drawn from the space of historically observed policy surprises, the approach does not require agents to form expectations about a new regime — the counterfactual is constructed as if the central bank had made a different sequence of choices within the prevailing framework.

This is the sense in which the method overcomes the Lucas critique: the IRFs from the prevailing rule remain valid as long as the counterfactual stays within the span of historically observed policy movements — that is, as long as the counterfactual represents a more aggressive or more restrained use of existing instruments, rather than

a fundamental regime change. We discuss the validity of this linearity assumption in more detail below (§4.1.2).

4.1.1 Core Equations

4.1.2 Assumptions and Their Validity in the Chinese Context

The CMW framework rests on several assumptions, drawn from McKay and Wolf (2023) and CMW (2025). Several of these are non-trivial in the Chinese institutional setting. We discuss each in turn, treating violations as scope conditions for our results rather than reasons to reject the approach outright.

Fixed fiscal rule. CMW assumes that fiscal policy follows a fixed rule throughout the sample, so that the estimated non-policy block is free of fiscal policy shocks. This assumption is clearly violated in China. The fiscal regime underwent substantial structural changes over our sample period: the abolition of the agricultural tax (2006), VAT reform, the rise and subsequent phase-out of local government financing vehicles (LGFVs), the transition from land-sale revenue dependence to centralized local government bond issuance, and periodic adjustments to central-local revenue and expenditure assignments. These changes mean that the estimated non-policy block may contain fiscal policy elements, potentially contaminating the impulse response estimates. The practical consequence is that our counterfactual captures the combined effect of monetary policy and any correlated fiscal adjustments, rather than monetary policy in isolation.

Unified fiscal authority. Relatedly, CMW assumes a single fiscal authority. In China, local governments account for approximately 80% of total government spending, and the degree of monetary-fiscal coordination varies across levels of government. Das and Song (2022) document that the State Council coordinates monetary and fiscal policy at the central level, but the extent to which local fiscal responses offset or amplify monetary policy transmission is difficult to estimate. This fragmentation introduces additional uncertainty into the impulse response estimates.

No information effects. McKay and Wolf (2023) require that monetary policy shocks contain no private information about fundamentals — that is, the IRFs must reflect pure policy transmission, not a signal about the central bank’s assessment of economic conditions. If shocks carry information content, the counterfactual would conflate transmission with signaling, and $\tilde{\nu}$ would be mis-identified. Whether PBoC actions contain significant information effects is an open question in the Chinese context. The Jarociński and Karadi (2020) sign-restriction approach — decomposing shocks into pure policy and information components based on the comovement of

interest rates and equity prices — would be the natural robustness check; we leave this for future work.

Symmetric effects of monetary policy. The framework assumes that contractionary and expansionary shocks of equal magnitude produce IRFs that are mirror images of each other. Visual inspection of sign-split IRFs — separating positive and negative FR007 shocks and estimating responses separately — suggests approximate symmetry in the shape of the responses.

Single policy instrument. CMW assumes that the central bank operates through a single instrument whose transmission can be captured by a single IRF. This is the most questionable assumption for the Chinese case, given the PBoC’s multi-instrument toolkit (§2.2). Our approach is to treat FR007 as a sufficient summary statistic for the overall policy stance: although the PBoC operates through multiple tools — OMOs, RRR adjustments, MLF, LPR, and formerly benchmark rates — all of these ultimately transmit through the interbank market and are reflected in FR007 movements. The counterfactual is therefore interpreted as: what would have happened if the PBoC had generated a different path for the short-term interbank rate, regardless of which specific instrument was used to achieve it. We provide a robustness check using the CMPI of Miranda-Agrippino, Nenova and Rey (2026, henceforth MANR) as an alternative policy indicator (§5.X).

4.2 Narrative Shock Identification

4.2.1 4.2.1 Genealogy

The Romer and Romer (2004) approach — regressing the policy rate on the central bank’s own forecasts and extracting residuals — does not translate directly to China for two reasons: the PBoC does not publish macroeconomic forecasts comparable to the Fed’s Greenbook, and there exist no authoritative estimates of potential output or the natural rate of interest for China. The Government Work Report (GWR) targets serve as a pragmatic substitute for these unobservable benchmarks.

Several papers have adapted the narrative approach to the Chinese context. Table 1 summarizes the key methodological choices across specifications.

	Chen, Xiao & Zha (2025)	Miranda-Agrippino et al. (2026)	This paper
LHS variable	FR007	CMPI	FR007
RHS timing	Contemporaneous	Lagged ($t - 1$)	Lagged ($t - 1$)
Exchange rate gap	No	Yes (CPR $\times$ spot-parity gap)	Yes

	Chen, Xiao & Zha (2025)	Miranda-Agrippino et al. (2026)	This paper
Asymmetric GDP response	Yes (threshold from CRZ 2018)	Yes (threshold $\approx$ 1pp)	Yes (threshold $\approx$ 1pp)
Sample	2002–2019	2002–2019	2002–2022

Chen, Ren and Zha (2018, henceforth CRZ), updated in Chen, Xiao and Zha (2025), estimated a monetary policy rule for China using the reserve requirement ratio (and later FR007) as the dependent variable, with an important innovation: an asymmetric, regime-switching response to the GDP gap, reflecting the PBoC’s greater tolerance for above-target growth than below-target growth.

MANR (2026) introduced the Composite Monetary Policy Indicator (CMPI) on the left-hand side of the regression, encompassing 36 policy instruments with time-varying equal weights (see Table A.1 in MANR). They further added an exchange rate gap term and used lagged $(t - 1)$ right-hand-side variables, which mechanically makes the policy rule forward-looking — though with the implicit assumption of perfect one-period foresight. MANR estimated the asymmetric GDP-gap threshold at approximately 1 percentage point, meaning that the PBoC tolerates small overshoots in output growth. Notably, only the coefficient on the negative GDP gap (below-target growth) is statistically significant — a finding we replicate in our FR007 specification. MANR also document a positive CPI response to contractionary monetary policy shocks, which they attribute to the user-cost methodology of Chinese housing CPI — a pattern we obtain under both narrative and HFI identification and discuss as a substantive finding in §5.X.

4.2.2 Our Specification

We combine elements from CXZ (2025) and MANR (2026). The monthly policy rule for the 7-day interbank repo rate (FR007) is:

$$\text{FR007}_t = \alpha + \rho \cdot \text{FR007}_{t-1} + \beta_\pi \cdot \text{cpi_gap}_{t-1} + \beta_y^+ \cdot \text{gap_pos}_{t-1} + \beta_y^- \cdot \text{gap_neg}_{t-1} + \beta_e \cdot \text{fx_gap}_{t-1} + \varepsilon_t^{\text{MP}}$$

where:

- $\text{cpi_gap}_t = \pi_t - \pi_t^*$, with π_t^* the annual CPI target from the Government Work Report (stepping in March at each NPC announcement, not January).
- $\text{gdp_gap}_t = g_t - g_t^*$, where g_t is a monthly real GDP proxy constructed via the Stock and Watson (2010) distribution method, and g_t^* is the GWR growth target.
- gap_pos_t , gap_neg_t : asymmetric split of the GDP gap around a 1pp threshold.
- fx_gap_t : a post-2006 indicator (after the end of the fixed exchange rate regime) interacting the change in the central parity rate with the spot–parity deviation.

The residual $\hat{\varepsilon}_t^{\text{MP}}$ is the narrative monetary policy shock series.

4.2.3 Why FR007 as the Policy Indicator

We retain FR007 rather than the CMPI on the left-hand side for two reasons. First, FR007 is the actual market rate that anchors the HFI surprises downstream (§4.3), maintaining internal consistency between the two shock identification strategies. Second, while the official 7-day reverse repo rate became the PBoC’s primary signaling tool only after 2015, the *market* FR007 has reflected interbank liquidity conditions and responded to all PBoC operations throughout the sample — including RRR adjustments and benchmark rate changes in the pre-2015 era. FR007 is not the policy instrument per se in the early sample; rather, it *transmits* whatever instrument the PBoC was using at the time.

Subsample estimation (2002–2015 vs. post-2015) yields qualitatively similar point estimates but with substantially wider confidence bands in the shorter post-2015 window, consistent with reduced effective sample size for a 6-lag monthly specification. The full-sample estimation is preferred for precision, and the FR007 transmission argument justifies pooling across policy-instrument regimes.

4.2.4 Critique of the CMPI Approach

While the CMPI solves the problem of an ever-changing policy toolkit by aggregating all instruments into a single index, three features limit its usefulness for our counterfactual exercise.

First, **interpretability**: the counterfactual shock $\tilde{\nu}$ applied to the CMPI lacks a transparent economic interpretation. A one-unit CMPI shock does not correspond to any specific policy action; it is a weighted average of deviations from trend across heterogeneous instruments. In contrast, a shock to FR007 maps directly onto an interbank rate movement.

Second, **cross-validation with HFI**: most PBoC policy tool changes are announced outside normal trading hours, so the market reaction to changes in individual tools cannot be cleanly isolated in daily data using a close-to-close HFI window. The CMPI, by construction, aggregates these tools and therefore cannot be externally validated by high-frequency market surprises in the way that FR007 can.

Third, **equal weights are atheoretical**: MANR assigns $\frac{1}{|N_t|}$ weight to each instrument active at date t , where $|N_t|$ is the number of active instruments. This treats a 25bp RRR cut — which directly affects trillions of yuan in required reserves — identically to a 10bp adjustment in an obscure short-term lending facility rate. There is no theoretical or empirical basis for this assumption. A potential improvement would

be to weight instruments by their associated liquidity impact or to use a factor model to extract a common policy component, but we leave this for future work.

4.3 High-Frequency Identification

4.3.1 Background

High-frequency identification of monetary policy shocks exploits the assumption that, within a narrow window around a policy announcement, movements in market interest rates are dominated by the surprise component of the policy action. This approach, developed for the US context by Kuttner (2001), Gürkaynak, Sack and Swanson (2005), and others, has been adapted to China by Das and Song (2022), who used daily changes in interest rate swaps (IRS) on FR007 around PBoC announcement dates. A distinct approach uses London copper prices as an outside instrument for Chinese monetary policy, exploiting China’s dominant role in global commodity demand.

4.3.2 Our HFI Specification

We identify monetary policy surprises as the daily close-to-close change in FR007 on dates when the PBoC announces a change to one of its core policy instruments:

$$\text{shock}_t = (\text{FR007}_t - \text{FR007}_{t-1}) \cdot \mathbf{1}_{t \in \mathcal{E}}$$

where $\mathcal{E}$ is the set of policy event dates. The core instruments tracked are:

- 7-day reverse repo rate (OMO) — primary short-term policy rate post-2016
- 1-year Medium-term Lending Facility (MLF) — introduced 2014
- 1-year Loan Prime Rate (LPR) — replaced benchmark lending rate in August 2019
- Reserve Requirement Ratio for large financial institutions (RRR)
- Benchmark deposit and lending rates (pre-2015, before rate liberalization)

Event dates are sourced from PBoC announcements collected via Wind (CMPI.xlsx). We restrict to dates on which a rate *change* occurs — dates with announcements confirming no change are excluded, as they generate minimal market surprise in the close-to-close window.

Daily shocks are summed to monthly frequency for use in the BVAR. This summation maintains a linearity assumption: transmission is additive across within-month shocks. If transmission is state-dependent, sign-asymmetric, or size-nonlinear, the monthly aggregation may introduce measurement error. We flag this as a maintained assumption rather than attempt to address it within the current framework.

4.3.3 Complementarity of Narrative and HFI Shocks

The two shock series serve distinct identification roles. The narrative shock is identification-by-construction: residuals from an estimated policy rule, capturing the

unsystematic component of interest rate movements after controlling for the PBoC’s reaction to inflation, growth, and exchange rate gaps. The HFI shock is identification-by-window: the market surprise within a narrow event window, with no assumption about the policy reaction function. The two approaches are complementary, not substitutes, and it is this complementarity that motivates using both as inputs to the counterfactual. In the counterfactual construction (§4.5), only the narrative shock series enters Π_m — the HFI serves as a robustness check on the estimated transmission mechanism.

4.3.4 Limitations and Potential Improvements

Several limitations of our HFI implementation merit acknowledgment.

IRS vs. raw FR007. Ideally, the HFI surprise would be measured using changes in the 1-year interest rate swap on FR007, which is more forward-looking and less affected by daily PBC liquidity operations than raw FR007. Das and Song (2022) use IRS for this reason. We use raw FR007 due to data access constraints (full IRS history requires a Wind terminal subscription).

Mid-month contamination. PBoC policy announcements — particularly MLF operations — tend to cluster around the 15th of each month, coinciding with NBS releases of major macroeconomic indicators (CPI, industrial production, etc.). This co-timing raises the concern that the HFI surprise reflects the market’s joint reaction to the policy action and the macro data release, rather than the policy surprise alone. Addressing this contamination rigorously would require intraday IRS data — minute-level snapshots around the policy announcement — which is available only for recent years and only to interbank market participants (Das and Song, 2022, provide validation for 2020 using Bloomberg tick data).

Two-stage HFI. The Miranda-Agrippino and Ricco (2021) two-stage approach — which first purges the high-frequency surprise of any predictable component due to the central bank’s information set — could further sharpen identification. We do not implement it here but note it as a potential extension.

Missing macroeconomic calendar controls. We do not formally control for the macroeconomic data release calendar in the HFI construction.

4.4 Impulse Response Estimation

The impulse responses used in both the transmission maps (Π matrices) and the robustness checks are estimated via a Bayesian SVAR with instrumental-variable identification. The narrative shock residuals $\hat{\varepsilon}_t^{\text{MP}}$ serve as the external instrument for the FR007 equation in the BVAR. For HFI-based robustness, the monthly-aggregated HFI series replaces the narrative residuals as the instrument.

The estimated IRFs exhibit a positive CPI response to contractionary monetary policy shocks — a price puzzle. We obtain this pattern under both narrative and HFI identification, consistent with the findings of MANR (2026). Rather than treating this as a misspecification, we interpret it as reflecting the user-cost methodology of the Chinese CPI’s housing component (居住类): Chinese residential mortgages carry floating rates that reprice immediately upon PBoC rate changes, creating a mechanical positive CPI response through the imputed homeowner cost channel. This interpretation follows Chodorow-Reich (2025). We discuss the implications for the counterfactual in §5.X.

4.5 Baseline Economic Forecast

The baseline forecast is constructed from a 7-variable Bayesian VAR with 6 lags estimated at monthly frequency. The variables are: monthly real GDP (constructed via Stock and Watson 2010 interpolation), CPI year-on-year, FR007, M2 year-on-year growth, RMB NEER year-on-year, US industrial production year-on-year, and industrial value added year-on-year. The estimation sample runs from 2002 to 2022 for the primary specification, with a 2002–2025 sample used as a robustness check. COVID dummies are included in all specifications.

We use a Minnesota-type prior following Giannone, Lenza and Primiceri (2015), as adopted by MANR (2026). The Minnesota prior addresses parameter proliferation — with 7 variables and 6 lags, the system has 43 regressors per equation — by shrinking coefficients toward a random-walk prior (own first lag = 1, all other coefficients = 0), with tightness declining harmonically in the lag order. This is the standard motivation for Bayesian shrinkage in richly parameterized macro VARs.

CMW (2025) use the language of the Wold representation to establish the theoretical validity of the counterfactual (Proposition 1). In practice, the computational object is simply the iterated BVAR forecast, which is equivalent to the Wold representation under invertibility. The baseline projection at horizon h from date t^* is:

$$\hat{y}_{t^*+h} = \hat{c} + \sum_{\ell=1}^6 \hat{A}_{\ell} \hat{y}_{t^*+h-\ell}, \quad h = 1, 2, \dots, H$$

with $H = 120$ months (10 years).

4.6 Counterfactual Construction

4.6.1 Setup

The counterfactual combines three objects from the preceding sections: baseline forecast paths $\pi^{\text{base}}, y^{\text{base}}, i^{\text{base}}$ over a T -month horizon; transmission maps Π_{π}, Π_y, Π_i linking

narrative shocks to each variable (constructed as Toeplitz matrices from the estimated IRFs); and policy targets π^*, y^* from the Government Work Report.

We solve for $\tilde{\nu} \in \mathbb{R}^T$, a sequence of narrative shocks — one per month. Counterfactual paths are the baseline plus the transmission:

$$\tilde{\pi} = \pi^{\text{base}} + \Pi_{\pi}\tilde{\nu}, \quad \tilde{y} = y^{\text{base}} + \Pi_y\tilde{\nu}, \quad \tilde{i} = i^{\text{base}} + \Pi_i\tilde{\nu}$$

Each Π matrix is $T \times T$ lower-triangular (Toeplitz), with the (t, s) entry giving the response of the variable at horizon $t - s$ to a unit shock at date s .

4.6.2 Loss Function

The shock sequence is chosen to minimize:

$$\mathcal{L}(\tilde{\nu}) = \lambda_{\pi}|\tilde{\pi} - \pi^*|^2 + \lambda_y|\tilde{y} - y^*|^2 + \lambda_i|\Delta\tilde{i}|^2 + \lambda_e|\tilde{e}|^2$$

The four terms penalize: (i) CPI deviation from the GWR inflation target; (ii) GDP deviation from the GWR growth target; (iii) abrupt changes in FR007 (interest-rate smoothing); and (iv) policy-induced exchange rate instability. The weights $\lambda_{\pi}, \lambda_y, \lambda_i, \lambda_e$ define the targeting rule. Following CMW (2025), the discount factor is set to 1. This formulation is algebraically equivalent to the rule-satisfaction minimization in Caravello et al. (2025), up to a sign convention on the shock vector.

We consider three targeting rules:

1. **CPI only:** $\lambda_{\pi} = 1, \lambda_y = 0, \lambda_i = 1, \lambda_e = 0$
2. **CPI + GDP:** $\lambda_{\pi} = 1, \lambda_y = 0.5, \lambda_i = 1, \lambda_e = 0$
3. **CPI + GDP + NEER:** $\lambda_{\pi} = 1, \lambda_y = 1, \lambda_i = 1, \lambda_e = 1$

4.6.3 Solution

Because the counterfactual paths are linear in $\tilde{\nu}$ and the loss is quadratic, the problem reduces to a weighted least-squares system:

$$\tilde{\nu}^* = \arg \min_{\tilde{\nu}} \mathcal{L}(\tilde{\nu}) = A_{\text{stack}} \setminus b_{\text{stack}}$$

where A_{stack} stacks the weighted transmission maps and b_{stack} stacks the weighted target gaps relative to the baseline. The solution is computed in closed form via a single linear solve.

4.6.4 Evaluation Against GWR Targets

We evaluate the counterfactual against the Government Work Report inflation targets rather than against the BVAR baseline. The baseline’s post-2023 forecasts inherit the deflationary trend that is itself the object of our analysis — measuring the counterfactual gap against the baseline would simply reproduce the baseline’s pessimism. The economically meaningful question is: by how much would more aggressive rate cuts

have closed the gap between actual (or counterfactual) inflation and the government’s stated target? This reframing makes the GWR targets the natural benchmark.

5 Data Construction

Note: This section is intended to slot between §4.3 (HFI) and §4.4 (Baseline Economic Forecast) in `methodology_draft.md`. The Stock-Watson construction details (§4.3bis.1) belong in a *data appendix*; the main text should summarize in one paragraph and point there. The BVAR variable list (§4.3bis.2) can be folded directly into the existing §4.4.

5.1 Monthly GDP Construction

Quarterly real GDP is the only available frequency from the NBS, but quarterly data produces unreliable narrative shock identification — CXZ (2025), who use quarterly frequency, obtain GDP responses exceeding 2% to a one-standard-deviation policy shock, far outside the plausible range. MANR (2026) use monthly data estimated by the New York Fed (Groen, 2020), which employs a dynamic factor model on dozens of indicator series. However, that series ends around 2020 and is therefore not suitable for our 2023-2025 counterfactual window.

We adopt the Stock and Watson (2010) distribution method, which allocates quarterly GDP to months using the adding-up identity $Q_T = q_{3T-2} + q_{3T-1} + q_{3T}$ and monthly indicator series that track the expenditure components. The method fits a state-space model with a Kalman smoother, ensuring that the monthly estimates sum exactly to the quarterly total reported by the NBS. [Reference: `stock_watson_distribute.py` implements this with a `break_year` option for sub-period estimation.]

Monthly indicators used for distribution:

- *Consumption:* Monthly total retail sales of consumer goods (社会消费品零售总额), in absolute value. Source: NBS via Wind.
- *Investment:* Urban fixed asset investment (城镇固定资产投资完成额). The overall FAI series is available only in growth rate form; we use the earliest available urban FAI in levels and backward-extend using the overall FAI growth rate. [TODO: Re-check exact start date and splicing logic in `Data_Monthly.ipynb` – confirm which series begins first and whether the backward calculation introduces any level shift.] Note that local government capital spending, which passes primarily

through FAI channels, is captured here — this partially alleviates the problem that the government spending series below covers only central government expenditure.

- *Government spending*: Monthly central government fiscal expenditure (全国一般公共预算支出). This series excludes local government expenditure and transfer payments. The omission of local spending is mitigated by its inclusion in FAI (above), but the omission of transfers remains a limitation.
- *Net exports*: Monthly merchandise trade balance (exports minus imports), recorded in USD. Converted to CNY using the monthly average CNY/USD spot rate constructed from daily Wind data. [Reference: akshare raw units are CNY per 100 USD pips; divide by 100 to match spot rate scale.]
- *CPI*: Monthly CPI index, used to deflate the constructed nominal monthly GDP into real terms.

The Stock-Watson method estimates a smooth monthly trend via cubic spline, detrends both the quarterly GDP and monthly indicators, estimates the parameters $(\beta, \rho, \sigma_\epsilon)$ by Gaussian maximum likelihood, and recovers the monthly path via the Kalman smoother. All input series are seasonally adjusted before distribution. [TODO: Move the full mathematical derivation (Equations 1–3 from Stock and Watson 2010) to Appendix X.]

Validation. [TODO: Report (i) correlation between constructed monthly GDP (summed to quarterly) and NBS quarterly real GDP; (ii) RMSE of the quarterly sum vs. reported quarterly GDP. The graph comparing the two series is necessary but not sufficient – include formal statistics. Reference: outputs/diagnostics/gdp_quarterly_vs_proxy_monthly.png.]

Robustness to GDP construction. Ferrière raised the concern that the composition of monthly indicators used in the Stock-Watson distribution may affect the estimated shocks and therefore the counterfactual results. A natural robustness check is to replace the constructed monthly GDP with industrial value added (IVA) — which is already in the BVAR as a separate variable and is the output measure used by MANR (2026) — and re-estimate the full pipeline. [TODO: Implement this robustness check: re-run BVAR and counterfactual with IVA replacing monthly GDP. If IRFs and counterfactual paths are qualitatively similar, report as robustness. If they diverge, discuss the source of divergence.]

5.2 BVAR Variables

Instruction: Fold the content below into the existing §4.4 of methodology_draft.md, replacing the one-sentence variable list.

The 7-variable BVAR is estimated at monthly frequency with the following variable ordering:

$$y_t = (g_t, IP_t, \pi_t, i_t, m_t, e_t, y_t^{\text{US}})'$$

Symbol	Variable	Transformation	Source
g_t	Real GDP	YoY growth (%)	Constructed via Stock-Watson (§4.3bis.1)
IP_t	Industrial Value Added (above designated scale)	YoY growth (%)	NBS
π_t	CPI	YoY (%)	NBS
i_t	FR007 (7-day inter- bank repo rate)	Level (%)	Wind
m_t	M2 money supply	YoY growth (%), entered as <code>pct_change(12) * 100</code>	PBoC
e_t	RMB Real Broad Effective Exchange Rate	YoY change (%)	BIS
y_t^{US}	US Industrial Production	YoY growth (%)	FRED

[TODO: Confirm M2 transformation matches current code – the earlier bug used `diff(12)` instead of `pct_change(12)`; counterfactual results were confirmed robust to this correction but the text should reflect the corrected specification.]

Variable selection rationale. The choice is driven by data availability and the need to avoid double-counting with the monthly GDP construction.

- *GDP and IVA both included:* GDP enters as the primary output measure for the counterfactual loss function; IVA enters as an independent monthly indicator of real activity that does not rely on the Stock-Watson construction. This is not double-counting — IVA is an NBS-reported series, while monthly GDP is a constructed proxy from expenditure components (consumption, investment, government spending, net exports). The two series capture overlapping but distinct dimensions of economic activity.
- *Consumption, FAI, net exports excluded from BVAR:* These expenditure components are already embedded in the monthly GDP construction. Including them separately would create mechanical correlation between the GDP proxy and its own inputs, inflating the apparent predictive content of the VAR.

- *Labor market variables excluded:* The registered unemployment rate (城镇登记失业率) is notoriously smoothed — it barely moved during the 2008–09 global financial crisis. The surveyed unemployment rate (城镇调查失业率) is available only from 2018. Other labor market indicators (job postings/searcher ratio, working hours, productivity) are either unavailable at monthly frequency or have insufficient sample length. CMW (2025) include labor market variables in their US specification; these are simply not available for China at comparable quality.
 - *US Industrial Production:* Controls for the external demand environment affecting Chinese growth. US IP is preferred over a US output gap measure because (i) it is directly observable (no unobservable potential output estimation required), (ii) it captures manufacturing-cycle spillovers to Chinese exports, and (iii) it is standard in the China VAR literature. [TODO: The slides say " y_t^{US} : US output gap YoY" but the code and methodology_draft.md say "US industrial production YoY." Reconcile – commit to US IP YoY and update slides.]
 - *M2:* Enters as an endogenous propagation channel capturing the quantity dimension of Chinese monetary policy, which was the dominant transmission mechanism in the pre-2015 era when interest rates were administratively repressed. Counterfactual results are robust to the M2 specification correction (diff vs. pct_change). [Confirmed in earlier analysis.]
 - *BIS Real Effective Exchange Rate:* Preferred over the CFETS nominal effective rate because (i) BIS REER is available for the full 2002–2025 sample, (ii) it adjusts for inflation differentials which matter for competitiveness and external transmission, and (iii) the CFETS index weights are only available from 2016.
 - *Year-on-year transformations:* Standard for monthly China macro work. Period-on-period rates would be better suited for structural papers isolating policy reaction timing but create inconsistency issues with the BVAR setup (mixing YoY and MoM generates cross-frequency dynamics that the VAR cannot cleanly handle).
-

5.3 Policy Targets from the Government Work Report

The GWR CPI and GDP growth targets define the counterfactual benchmarks (π_t^* , g_t^*). Targets are extracted from the annual Government Work Report and, for earlier years, the corresponding Five-Year Plans. Several coding decisions merit documentation:

- *Timing:* GWR targets step in March (the NPC announcement month), not January. This is consequential for the 2025 target change in the narrative shock construction — the CPI target was lowered to “around 2%” at the March 2025 NPC session (following the CEWC signal in December 2024).
- *Ranges:* When the GWR states a range (e.g., “7–7.5%”), we take the midpoint.

- *2020 (no target)*: The government issued no GDP growth target for 2020 due to COVID. We test both dropping 2020 from the target series and imputing the 2019 target value; the counterfactual results show negligible sensitivity to this choice.
- *Exchange rate target*: For the narrative policy rule (§4.2.2), the exchange rate gap uses only the CNY/USD bilateral rate, not the full CFETS basket. Two reasons: (i) the USD rate is unambiguously the most important reference point for PBoC FX policy; (ii) the CFETS basket weights used by MANR (2026) are available only from 2016, making full-sample estimation infeasible.

[TODO: Create an appendix table listing all GWR targets (CPI and GDP) by year, 2002–2025, with the NPC announcement date and any notes on range-to-midpoint conversion.]

5.4 CMPI Reconstruction

The Composite Monetary Policy Indicator (CMPI) of Miranda-Agrippino, Nenova and Rey (2026) aggregates deviations from trend across the PBoC’s full instrument toolkit into a single index. We reconstruct the CMPI series extending the sample from 2002 to 2025, using the same 13 instrument categories documented in MANR Table A.1 (sourced from CMPI.xlsx via Wind). [TODO: List the 13 instrument categories explicitly – the full list can be found in the CMPI.xlsx sheet names.]

The CMPI is used only for robustness; the main counterfactual uses narrative shocks identified on FR007 (§4.2.3). The reason is that the counterfactual requires shocks on a variable with transparent economic interpretation — a one-unit FR007 shock maps to a basis-point interbank rate movement, while a one-unit CMPI shock is a weighted average across heterogeneous instruments with no direct policy interpretation (§4.2.4). The CMPI enters the BVAR in the robustness specification as an additional variable, with a separate policy rule estimated on CMPI instead of FR007 (see `RRShocks_monthly_cmpi.jl`).

6 Appendix items to produce

Checklist of appendix content implied by the sections above.

1. *Appendix A.X: Stock-Watson Monthly GDP Construction*
 - Full mathematical derivation (Equations 1–3 from Stock and Watson 2010 memorandum)
 - Description of trend estimation (cubic spline), Kalman smoother, parameter estimation (Gaussian MLE)

- Seasonal adjustment procedure
 - Break-year option and sub-period estimation
 - Comparison figure: constructed monthly GDP vs. NBS quarterly GDP (already at `outputs/diagnostics/gdp_quarterly_vs_proxy_monthly.png`)
 - [TODO: Add formal comparison statistics: correlation, RMSE]
2. *Appendix A.X: GWR Target Table*
 - Year, CPI target, GDP target, NPC date, notes (range → midpoint, 2020 no-target handling)
 3. *Appendix A.X: CMPI Instrument List*
 - All 13 instrument sheets from CMPI.xlsx with date coverage
 4. *Appendix A.X: Robustness — IVA replacing monthly GDP*
 - [TODO: Run and report]
 5. *Appendix A.X: Data sources summary table*
 - Variable, source, frequency, sample period, transformation
-

7 Outstanding TODOs (consolidated)

Priority	Item	Location
<i>High</i>	Report correlation and RMSE for monthly GDP vs. quarterly GDP	§4.3bis.1 / Appendix
<i>High</i>	Reconcile US variable: slides say “output gap YoY,” code/text says “US IP YoY” — commit to US IP YoY and update slides	§4.3bis.2 / slides
<i>High</i>	Confirm M2 enters as <code>pct_change(12)*100</code> in current code	§4.3bis.2 / §4.4
<i>Medium</i>	Re-check FAI backward calculation: exact start date, splicing logic, potential level shift	§4.3bis.1
<i>Medium</i>	Implement Ferrière robustness: re-run BVAR + counterfactual with IVA replacing monthly GDP	§4.3bis.1 / Appendix
<i>Medium</i>	Create appendix table of GWR targets 2002–2025	§4.3bis.3
<i>Medium</i>	List all 13 CMPI instrument categories explicitly	§4.3bis.4
<i>Low</i>	Move Stock-Watson math to appendix (full derivation)	§4.3bis.1 → Appendix
<i>Low</i>	Regime-dependent policy response (Zha-style threshold across GDP gap sign) — flag as future work, do not scope-creep before May 17	Notes
<i>Low</i>	Create appendix data sources summary table	Appendix

8 Results

8.1 Impulse Responses

8.1.1 Narrative Identification

Figure X reports impulse responses of monthly GDP, CPI, NEER, and industrial production (IP) to a 1 percentage point contractionary shock to FR007, identified using the narrative residuals from the policy rule (§4.2). The shock is normalized so that FR007 rises by 1pp on impact.

Real GDP exhibits a brief positive spike of approximately 0.25pp in the first month before declining to zero around the fourth month, reaching a trough of roughly -0.5 pp near the tenth month, and recovering gradually thereafter. The initial positive response is consistent with two (non-exclusive) interpretations: a residual information effect — the rate hike signals confidence in underlying growth — and the mechanical possibility, at monthly frequency, that output is still expanding in the month when the PBoC tightens. The slow recovery beyond the trough is worth noting; we return to this persistence in the subsample analysis (Appendix X).

CPI rises to a peak of approximately 0.25pp around the fifth month before declining through zero near the fourteenth month and reaching a trough at roughly -0.125 pp. This positive CPI response to a contractionary shock — a price puzzle — mirrors the finding in Miranda-Agrippino, Nenova and Rey (2026, henceforth MANR), who report a peak of approximately 0.5pp with a zero crossing around the twelfth month using CMPI-based identification. We interpret the price puzzle not as a misspecification but as reflecting the user-cost methodology of the Chinese CPI’s housing component (居住类). Chinese residential mortgages carry floating rates that reprice immediately upon PBoC rate changes, creating a mechanical positive CPI response through the imputed homeowner cost channel (Chodorow-Reich, 2025). This interpretation is corroborated by the CMPI-based IRFs (Appendix X), where the use of a broader policy indicator that captures non-rate instruments — which do not directly affect mortgage costs — yields a substantially smaller initial CPI increase and a more conventional negative response within five months. [TODO: Verify NBS 自有住房 CPI sub-component methodology over 2003–2025.]

The real effective exchange rate appreciates, consistent with standard theory: the NEER rises to a peak of approximately 1.25pp around the tenth month before reverting toward zero. Industrial production, included both as a robustness check on the GDP response and as an input to the counterfactual, displays a qualitatively similar pattern — an initial spike to 0.5pp around the third month, a zero crossing at the fourth month, and a trough near -1 pp around the thirteenth month. The IP response is roughly

double the GDP response in magnitude, which likely reflects the higher cyclical sensitivity of industrial output relative to services-inclusive GDP in the Chinese economy.

8.1.2 High-Frequency Identification

Figure X reports the corresponding IRFs using the rate-change-only HFI shock series (§4.3). The medium-run dynamics are broadly consistent with the narrative identification: GDP reaches a trough of approximately -0.5pp near the tenth month, CPI peaks at 0.6pp around the fifth month and crosses zero near the sixteenth month, and the NEER peaks at 1.75pp around the tenth month. The principal discrepancy is in the short run. GDP and IP exhibit substantially larger initial spikes (approximately 2pp and 1.5pp , respectively) under HFI identification, which we attribute to known measurement issues in the monthly HFI series: contamination from the overlap between PBoC announcement dates and NBS macro data releases (approximately one-fifth of rate-change dates coincide with major macro releases; see §4.3), and the noise introduced by aggregating daily close-to-close changes to monthly frequency.

The convergence of medium-run responses across two independent identification strategies — one exploiting the systematic policy rule, the other exploiting announcement-day market movements — strengthens the case for the narrative identification as the primary input to the counterfactual. The HFI serves as robustness: the later transmission pattern is confirmed, while the short-run discrepancies are traceable to well-understood measurement limitations. [TOD0: Potential improvements include using intraday IRS data on FR007 (if available via Wind) and a two-stage HFI approach that separates the information effect.]

8.1.3 Subsample Comparison

[TOD0: Move subsample IRF comparison (2002–2015 benchmark-rate era / 2015–2025 FR007-corridor era) from appendix to main body. Discuss as a substantive finding: structural break in PBoC operating framework is reflected in the transmission mechanism.]

8.2 Baseline Forecast

The counterfactual requires a no-policy-change baseline — the BVAR’s iterated forecast from the end of the estimation sample. We estimate the BVAR through 2022 and generate recursive forecasts for 2023–2025.

GDP and IP forecasts track realized values well. The FR007 forecast stabilizes around 2.5%, substantially above the declining realized path toward approximately 1.5% by end-2025 — reflecting the BVAR’s inability to anticipate the post-2022 easing

cycle, which is precisely the policy variation the counterfactual is designed to evaluate. The NEER forecast exhibits a similar upward drift relative to realized values.

The CPI forecast merits particular attention. The BVAR projects CPI of approximately 1.75% over 2023–2025, against realized values near zero. Even when the BVAR is re-estimated through 2025, the out-of-sample CPI forecast settles around 1%, still well above realized inflation. This persistent overprediction is not surprising — the BVAR is estimated on a sample in which CPI rarely approached zero and never sustained a prolonged period of near-deflation — but it has a direct consequence for the counterfactual. Because the baseline already expects CPI well above zero, measuring the counterfactual gap against the baseline would understate the policy challenge: the relevant question is not how much additional inflation the counterfactual generates relative to an already-optimistic baseline, but how close the counterfactual brings CPI to the government’s stated target. This motivates the evaluation against GWR targets rather than against the BVAR baseline (§4.5.4).

[Figure: Monthly GDP, CPI, FR007, NEER – BVAR forecast (2022 sample, recursive 2023–2025) vs. realized values.]

8.3 5.3 Counterfactual

We present counterfactual results for three targeting specifications, ordered from the most to the least comprehensive in terms of the PBoC’s objective set. The baseline throughout is FR007 smoothing ($\lambda_i = 1$), which penalizes abrupt rate changes and prevents the optimizer from producing implausible rate paths. [TODO: Add credible intervals for counterfactual paths.]

8.3.1 Full Specification: CPI + GDP + NEER

The full specification ($\lambda_\pi = 1, \lambda_y = 1, \lambda_i = 1, \lambda_e = 1$) is the most policy-relevant scenario, as it approximates the PBoC’s revealed multi-objective mandate — price stability, output growth, and exchange rate management. Under this specification:

- GDP tracks slightly above the BVAR forecast and realized values.
- FR007 and the NEER largely follow the baseline forecast: FR007 increases relative to the realized path, and the NEER appreciates modestly above realized values for some periods. Because the FR007 deviation from the baseline is small, IP is essentially unaffected.
- CPI rises only slightly above the BVAR forecast, settling around 2% under the 2022 sample — and around 1.3% under the 2025 sample.

This is the central finding of the paper. When the counterfactual optimizer is tasked with simultaneously closing the CPI gap, stabilizing GDP near target, and limiting

exchange rate volatility, the resulting policy path does not differ dramatically from what the PBoC actually implemented. The implication is that, conditional on the estimated transmission mechanism and the multi-objective loss function, the PBoC’s realized policy was approximately efficient: the 2023–2025 inflation undershoot reflects the binding constraint of the multi-objective mandate rather than a failure of policy optimization. More aggressive rate cuts could have raised CPI closer to target, but only at the cost of exchange rate instability or output overshooting — costs that the full-specification loss function internalizes.

8.3.2 CPI + GDP

Relaxing the exchange rate objective ($\lambda_e = 0$) while retaining the GDP target reveals the tradeoff structure. GDP now tracks close to target and to realized values — the GDP constraint is not binding in any specification, consistent with the PBoC’s effective prioritization of growth. CPI becomes more smoothed, falling slightly below the GWR target for some periods. The key difference appears in FR007: the counterfactual rate path rises to 3–3.5%, substantially above both the BVAR forecast and realized values. This generates a correspondingly larger NEER appreciation and a decline in IP from the 5% path to approximately 3%.

The economic interpretation is straightforward: without the exchange rate constraint, the optimizer exploits the CPI price puzzle — higher rates raise CPI through the housing cost channel in the short run — but the cost is borne by the external sector. The fact that removing the FX objective produces such different rate and exchange rate paths underscores the extent to which the exchange rate constraint binds in the full specification.

8.3.3 CPI Only

The most restrictive specification ($\lambda_y = 0, \lambda_e = 0$) targets only CPI deviation from the GWR target, subject to rate smoothing. Here, FR007 needs to be lower than the BVAR forecast (which projects rates above 2.25%) but remains higher than the realized path. GDP overshoots to approximately 7%. The NEER appreciates above the baseline. IP remains close to both the forecast and the actual path — because the FR007 deviation from the baseline forecast is relatively small under this specification, the transmission into IP is muted.

This specification isolates the partial-equilibrium effect of CPI targeting: the optimizer generates modest rate cuts that would have kept CPI somewhat closer to target, but without accounting for the output and exchange rate consequences. The fact that even pure CPI targeting does not require dramatically different rate paths from those observed reinforces the finding from the full specification — the binding constraints on

PBoC policy lie in the multi-objective structure, not in a failure to respond to inflation per se.

8.3.4 Discussion

The progression from the full specification to the CPI-only case reveals a clear hierarchy of binding constraints. The exchange rate constraint is the tightest: removing it changes the counterfactual rate path by 1-1.5pp and produces qualitatively different NEER dynamics. The GDP constraint is effectively slack — growth was near target in 2023-2025, and the optimizer does not need to trade off CPI against output. The rate-smoothing penalty prevents all specifications from producing implausible rate volatility.

A post-2023 caveat applies to all three specifications. The BVAR baseline inherits the sample’s central tendency — inflation averaging well above zero — and therefore forecasts CPI around 1.75% even as realized CPI collapsed. The counterfactual CPI paths are thus computed relative to an already-elevated baseline, which means the *level* of counterfactual CPI should be read with some caution. The economically meaningful object is the gap between the counterfactual and the GWR target, not the absolute level: this gap is small under the full specification, confirming that the PBoC’s realized policy was approximately consistent with its stated objectives given the multi-instrument, multi-objective environment it operates in.

[TODO: Sensitivity analysis – varying λ weights. Currently equal weights produce similar results across reasonable perturbations; note this and cross-reference appendix.]

8.4 Robustness

8.4.1 CMPI-Based Counterfactuals

[TODO: Report counterfactual results using CMPI shocks (MANR 2026) as the transmission input. Note interpretation difficulties: a 1pp shock to CMPI does not have the same policy interpretation as a 1pp shock to FR007, and the CMPI’s equal weighting across instruments is atheoretical. Cross-reference CMPI IRFs in appendix.]

8.4.2 Full-Sample Estimation

[TODO: Report counterfactual results using BVAR and IRFs estimated through 2025 rather than 2022. The main difference is in the baseline forecast (lower CPI projection); IRFs are similar across sample endpoints. Discuss implications for the post-2023 caveat.]

[TODO: Consider whether the 3-year forecast horizon is a limitation. CMW (2025) use approximately 5 years; shorter horizons may understate cumulative transmission.]

9 Discussion and Limitations

10 Conclusion

11 Appendix

11.1 A.1 Data Construction

11.1.1 A.1.1 Monthly GDP via Stock-Watson Distribution

Quarterly real GDP is the only available frequency from the NBS, but quarterly data produces unreliable narrative shock identification — CXZ (2025), who use quarterly frequency, obtain GDP responses exceeding 2% to a one-standard-deviation policy shock, far outside the plausible range. MANR (2026) use monthly data estimated by the New York Fed (Groen, 2020), which employs a dynamic factor model on dozens of indicator series. However, that series ends around 2020 and is therefore not suitable for our 2023-2025 counterfactual window.

We adopt the Stock and Watson (2010) distribution method, which allocates quarterly GDP to months using the adding-up identity $Q_T = q_{3T-2} + q_{3T-1} + q_{3T}$ and monthly indicator series that track the expenditure components. The method fits a state-space model with a Kalman smoother, ensuring that the monthly estimates sum exactly to the quarterly total reported by the NBS.

Monthly indicators used for distribution:

Component	Series	Notes	Source
Consumption	Total retail sales of consumer goods (社会消费品零售总额)	Absolute value	NBS via Wind
Investment	Urban fixed asset investment (城镇固定资产投资完成额)	Backward-extended using overall FAI growth rate for pre-availability period. Captures local government capital spending, partially offsetting the central-only fiscal expenditure series below.	NBS via Wind

Component	Series	Notes	Source
Government spending	Central government fiscal expenditure (全国一般公共预算支出)	Excludes local government expenditure and transfer payments	MoF via Wind
Net exports	Merchandise trade balance (exports – imports)	Recorded in USD; converted to CNY using monthly average CNY/USD spot rate from daily Wind data	GAC via Wind
Deflator	Monthly CPI index	Used to deflate constructed nominal monthly GDP to real terms	NBS

[TODO: Re-check exact start date and splicing logic for urban FAI backward calculation in Data_Monthly.ipynb – confirm whether the backward extension introduces any level shift.]

The Stock-Watson method estimates a smooth monthly trend via cubic spline, detrends both the quarterly GDP and monthly indicators, estimates the parameters $(\beta, \rho, \sigma_\epsilon)$ by Gaussian maximum likelihood, and recovers the monthly path via the Kalman smoother. All input series are seasonally adjusted before distribution.

[TODO: Report formal validation statistics: (i) correlation between constructed monthly GDP (summed to quarterly) and NBS quarterly real GDP; (ii) RMSE of the quarterly sum vs. reported quarterly GDP. Reference: outputs/diagnostics/gdp_quarterly_vs_proxy_monthly.png.]

11.1.2 A.1.2 BVAR Variable Definitions

The 7-variable BVAR is estimated at monthly frequency with the following variable ordering:

$$y_t = (g_t, IP_t, \pi_t, i_t, m_t, e_t, y_t^{\text{US}})'$$

Symbol	Variable	Transformation	Source
g_t	Real GDP	YoY growth (%)	Constructed via Stock-Watson (§A.1.1)
IP_t	Industrial Value Added (above designated scale)	YoY growth (%)	NBS

Symbol	Variable	Transformation	Source
π_t	CPI	YoY (%)	NBS
i_t	FR007 (7-day inter-bank repo rate)	Level (%)	Wind
m_t	M2 money supply	YoY growth (%), entered as <code>pct_change(12) * 100</code>	PBoC
e_t	RMB Real Broad Effective Exchange Rate	YoY change (%)	BIS
y_t^{US}	US Industrial Production	YoY growth (%)	FRED

Variable selection rationale.

GDP and IVA both included. GDP enters as the primary output measure for the counterfactual loss function; IVA enters as an independent monthly indicator of real activity that does not rely on the Stock-Watson construction. This is not double-counting — IVA is an NBS-reported series, while monthly GDP is a constructed proxy from expenditure components.

Consumption, FAI, net exports excluded from BVAR. These expenditure components are already embedded in the monthly GDP construction. Including them separately would create mechanical correlation between the GDP proxy and its own inputs.

Labor market variables excluded. The registered unemployment rate (城镇登记失业率) is notoriously smoothed — it barely moved during the 2008–09 global financial crisis. The surveyed unemployment rate (城镇调查失业率) is available only from 2018. Other labor market indicators lack sufficient sample length. CMW (2025) include labor market variables in their US specification; these are simply not available for China at comparable quality.

US Industrial Production. Controls for the external demand environment. Preferred over a US output gap measure because it is directly observable, captures manufacturing-cycle spillovers to Chinese exports, and is standard in the China VAR literature.

M2. Enters as an endogenous propagation channel capturing the quantity dimension of Chinese monetary policy, which was the dominant transmission mechanism in the pre-2015 era of interest rate repression. Counterfactual results are robust to the M2 specification correction (diff vs. pct_change).

BIS Real Effective Exchange Rate. Preferred over the CFETS nominal effective rate because (i) BIS REER is available for the full 2002–2025 sample, (ii) it adjusts for inflation differentials, and (iii) CFETS index weights are only available from 2016.

Year-on-year transformations. Standard for monthly China macro work. Period-on-period rates would be better suited for structural papers isolating policy reaction timing but create inconsistency issues with the BVAR setup.

11.1.3 A.1.3 Government Work Report Targets

The GWR CPI and GDP growth targets define the counterfactual benchmarks (π_t^* , g_t^*). Targets step in March at each NPC announcement, not January. When the GWR states a range (e.g., “7–7.5%”), we take the midpoint. The government issued no GDP growth target for 2020; we test both dropping 2020 and imputing the 2019 target value, with negligible sensitivity in the counterfactual.

[TOD0: Insert table – Year, CPI target (%), GDP target (%), NPC date, notes.]

For the narrative policy rule (§4.2.2), the exchange rate gap uses only the CNY/USD bilateral rate rather than the full CFETS basket, because (i) the USD rate is unambiguously the most important reference point for PBoC FX policy and (ii) CFETS basket weights are available only from 2016.

11.1.4 A.1.4 CMPI Reconstruction

The Composite Monetary Policy Indicator of Miranda-Agrippino, Nenova and Rey (2026) aggregates deviations from trend across the PBoC’s full instrument toolkit into a single index. We reconstruct the CMPI series from 2002 to 2025 using the 13 instrument categories documented in MANR Table A.1 (sourced from CMPI.xlsx via Wind).

[TOD0: List the 13 instrument categories explicitly with date coverage for each sheet.]

The CMPI is used only for robustness; the main counterfactual uses narrative shocks identified on FR007 (§4.2). The reason is that the counterfactual requires shocks on a variable with transparent economic interpretation — a one-unit FR007 shock maps to a basis-point interbank rate movement, while a one-unit CMPI shock is a weighted average across heterogeneous instruments with no direct policy interpretation.

11.1.5 A.1.5 Data Sources Summary

[TOD0: Insert comprehensive table – Variable, Source, Frequency, Sample period, Transformation, Wind code (where applicable).]

11.2 A.2 Narrative Shock Identification: Supplementary Results

11.2.1 A.2.1 Policy Rule Regression Results

Table A.2 reports the estimated coefficients from the FR007 policy rule (§4.2.2) for three samples: full sample (2002–2022), the benchmark-rate era (2002–2015), and the FR007-corridor era (2015–2022). The coefficients on `cpi_gap` and `gap_neg` are statistically significant across specifications, consistent with MANR (2026). The asymmetric GDP gap structure — in which the PBoC responds to below-target growth but tolerates above-target growth — is confirmed: $\hat{\beta}_y^-$ is significant while $\hat{\beta}_y^+$ is not.

[Figure: Combined regression table – full sample, 2002–2015, 2015–2022.]

11.2.2 A.2.2 Subsample IRF Comparison

Figure A.X compares IRFs estimated on the 2002–2015 and 2015–2025 subsamples. The 2002–2015 estimates exhibit tighter credible bands, consistent with the longer effective sample. The 2015–2025 estimates are qualitatively similar in shape but noisier, particularly for GDP and IP. The CPI price puzzle is present in both subsamples, supporting the user-cost housing channel interpretation — the mortgage rate mechanism operates throughout the sample, irrespective of the PBoC’s shifting instrument mix.

[Figure: Side-by-side IRFs, 2002–2015 vs. 2015–2025, with 68% and 90% credible bands.]

11.2.3 A.2.3 Sign Asymmetry

Figure A.X presents IRFs estimated separately for positive (contractionary) and negative (expansionary) narrative shocks. The shapes are approximately mirror images, consistent with the linearity assumption maintained by the CMW framework (§4.1.2). A visual comparison suggests approximate symmetry, though formal Wald-type testing would be required to confirm this statistically.

[Figure: IRFs for positive vs. negative narrative shocks, all BVAR variables.]

[TODO: Consider implementing a formal symmetry test if time permits before May 17.]

11.3 A.3 HFI Shock Identification: Supplementary Results

11.3.1 A.3.1 Rate-Change-Only vs. All-Announcement Shocks

Figure A.X compares IRFs using the rate-change-only HFI series against the all-announcement series (which includes dates where the PBoC confirmed no change). The rate-change-only series produces substantially cleaner IRFs — the all-announcement series introduces noise from no-change dates where the close-to-close FR007 movement

reflects liquidity conditions rather than policy surprises. This motivates our choice of the rate-change-only series for robustness.

[Figure: IRFs from rate-change-only vs. all-announcement HFI, all BVAR variables.]

11.3.2 A.3.2 HFI Subsample Comparison

Figure A.X compares HFI-based IRFs across the 2002–2015 and 2015–2025 subsamples. As with the narrative identification, the later subsample produces wider credible bands. The initial GDP and IP spikes that characterize the full-sample HFI results (§5.1.2) are more pronounced in the post-2015 subsample, consistent with the hypothesis that mid-month contamination from concurrent NBS macro data releases is a larger problem in the recent period when PBoC operations have become more frequent and more closely scheduled around the 15th.

[Figure: HFI IRFs, 2002–2015 vs. 2015–2025.]

11.3.3 A.3.3 HFI Sign Asymmetry

Figure A.X presents HFI-based IRFs estimated separately for positive and negative shocks. The results are noisier than the narrative sign-split exercise (§A.2.3) but qualitatively consistent with approximate symmetry.

[Figure: HFI IRFs for positive vs. negative shocks.]

11.3.4 A.3.4 HFI Shock Series

Figure A.X plots the monthly-aggregated rate-change-only HFI shock series over 2002–2025. Major episodes — the 2008 easing, 2010–11 tightening, 2015 easing, and 2022–2024 easing — are visible as clusters of same-sign shocks.

[Figure: Time series of monthly HFI shocks.]

11.4 A.4 CMPI-Based Identification: Supplementary Results

11.4.1 A.4.1 CMPI Policy Rule Regression

Table A.X reports the estimated coefficients from the CMPI policy rule (replacing FR007 with CMPI as the dependent variable). The regression is reported for three samples as in §A.2.1.

[Figure: CMPI regression table – full sample, 2002–2015, 2015–2022.]

[TODO: Consider adding a version restricting to important-instruments-only dates.]

11.4.2 A.4.2 CMPI Shock Series

Figure A.X plots the CMPI narrative shock series alongside the FR007 narrative shock series for comparison. The two series are positively correlated but not identical, reflecting the CMPI’s inclusion of non-rate instruments.

[Figure: CMPI vs. FR007 narrative shock series.]

11.4.3 A.4.3 CMPI IRFs

Figure A.X reports IRFs to a 1pp CMPI shock for the 2002–2022 sample. The key differences from the FR007-based IRFs (§5.1.1) are:

- *GDP*: No initial positive spike; slightly negative in the first period, reaching a trough of approximately -1.5 pp around the eighth month before recovering. The absence of the initial spike is consistent with the CMPI capturing non-rate instruments that do not generate the same information-effect or monthly-frequency GDP-hike dynamics.
- *CPI*: A much smaller initial increase (peak 0.2 pp), crossing zero by the fifth month and reaching a trough of -0.8 pp around the twelfth month. This substantially reduced price puzzle corroborates the user-cost housing channel interpretation: the CMPI includes non-rate instruments (RRR, structural tools) that do not directly affect mortgage costs, attenuating the mechanical positive CPI response.
- *FR007*: Very wide credible bands, which is expected — FR007 is only one of the instruments entering the CMPI, and a composite shock need not move the interbank rate consistently.
- *NEER*: Appreciates to approximately 2pp around the eighth month, qualitatively similar to the FR007-based response.
- *IP*: Starts negative, reaches a trough of approximately -3 pp around the eighth month, roughly triple the FR007-based IP response. This amplification may reflect the CMPI’s inclusion of RRR changes, which directly affect bank lending capacity and thus investment-heavy industrial output.

The overall pattern suggests that accounting for the full instrument toolkit produces more conventional-looking IRFs, particularly for CPI. The equal-weight assumption remains a caveat (§4.2.4).

[TODO: Note that some CMPI values are slightly negative – consider whether to impose non-negativity.]

11.4.4 A.4.4 CMPI Subsample and Sign-Asymmetry IRFs

[Figure: CMPI IRFs, 2002–2015 vs. 2015–2025.]

[Figure: CMPI sign-split IRFs.]

11.5 A.5 Baseline Forecast Diagnostics

11.5.1 A.5.1 BVAR Residuals and Forecast Decay

Figure A.X plots the BVAR residuals for each equation over the estimation sample. Figure A.Y plots the forecast root mean squared error as a function of forecast horizon (1 to 36 months) for each variable, illustrating the rate at which forecast precision deteriorates.

[Figure: Residual time series for each BVAR equation.]

[Figure: Forecast RMSE by horizon.]

11.5.2 A.5.2 GDP Recursive Forecast: 2025 Sample

Figure A.X reports the recursive GDP forecast from the BVAR estimated through 2025, compared against the 2022-sample forecast and realized values. The 2025-sample forecast benefits from incorporating the post-2022 easing cycle but still overpredicts CPI.

[Figure: GDP forecast comparison – 2022 sample vs. 2025 sample vs. realized.]

11.5.3 A.5.3 CPI Baseline Forecast Comparison

Figure A.X compares CPI forecasts from two BVAR specifications: (i) estimated through 2022 with recursive 2023–2025 forecasts (main specification), and (ii) estimated through 2025 with out-of-sample forecasts. The 2022-sample BVAR projects CPI of approximately 1.75%; the 2025-sample BVAR projects approximately 1%. Both overshoot realized CPI (0%), confirming the structural break interpretation: the BVAR cannot generate near-zero CPI from within-sample dynamics. This motivates the GWR-target evaluation benchmark (§4.5.4, §5.3.4).

[Figure: CPI forecasts – 2022 sample, 2025 sample, realized, GWR target.]

11.6 A.6 Counterfactual: Supplementary Results

11.6.1 A.6.1 Full-Sample Estimation (2025 Sample)

Figure A.X reports counterfactual results for all three targeting specifications using the BVAR and IRFs estimated through 2025 rather than 2022. The primary difference lies in the baseline forecast — lower projected CPI and FR007, closer to the realized easing path. The IRFs are qualitatively similar across sample endpoints. Under the full specification (CPI + GDP + NEER), counterfactual CPI settles around 1.3% (vs. 2% with the 2022 sample), reflecting the more pessimistic baseline. The qualitative finding — that the PBoC’s realized policy was approximately efficient given the multi-objective mandate — is preserved.

[Figure: Counterfactual paths, all three specifications, 2025 sample.]

11.6.2 A.6.2 CMPI-Based Counterfactual

Figure A.X reports counterfactual results using CMPI shocks as the transmission input, replacing the narrative FR007 shocks. Interpretation requires caution: a unit CMPI shock does not have the same policy meaning as a unit FR007 shock (§4.2.4), and the transmission maps are constructed from CMPI IRFs (§A.4.3) rather than FR007 IRFs.

[TODO: Report results. If qualitatively similar to the FR007-based counterfactual, this supports robustness; if they diverge, discuss the source – likely the reduced price puzzle in CMPI IRFs and the amplified IP response.]

[TODO: The counterfactual with CMPI may need a separate sample/transmission matrix to accommodate the different shock interpretation. Document the construction.]

11.6.3 A.6.3 Sensitivity to Loss Function Weights

Figure A.X reports counterfactual CPI and FR007 paths under perturbations of the loss function weights $(\lambda_\pi, \lambda_y, \lambda_i, \lambda_e)$ around the baseline equal-weight specification. All perturbations within a reasonable range (halving or doubling individual weights) produce qualitatively similar counterfactual paths, motivating the choice of equal weights in the main specification.

[Figure: Counterfactual CPI and FR007 paths under weight perturbations.]

[TODO: Consider whether this belongs in the main body (§5.3) rather than the appendix, since equal-weight insensitivity is a substantive finding.]

11.7 A.7 Outstanding TODOs (Consolidated)

Priority	Item	Section
<i>High</i>	Credible intervals for counterfactual paths	§5.3
<i>High</i>	Formal validation statistics for monthly GDP (correlation, RMSE vs. quarterly NBS)	§A.1.1
<i>High</i>	Reconcile US variable label on slides vs. text — commit to US IP YoY	§A.1.2 / slides
<i>High</i>	Verify NBS 自有住房 CPI sub-component methodology 2003–2025	§5.1.1 / §A.4.3
<i>Medium</i>	GWR target table (all years 2002–2025)	§A.1.3
<i>Medium</i>	List 13 CMPI instrument categories with date coverage	§A.1.4
<i>Medium</i>	Data sources summary table	§A.1.5
<i>Medium</i>	Re-check FAI backward calculation splicing logic	§A.1.1
<i>Medium</i>	Ferrière robustness: BVAR + counterfactual with IVA replacing monthly GDP	§A.1.1 / §A.6
<i>Medium</i>	Tabulate policy-event / macro-release date overlap	§A.3.2

Priority	Item	Section
<i>Medium</i>	CMPI non-negativity constraint consideration	§A.4.3
<i>Low</i>	Formal symmetry test (Wald-type) for sign-split IRFs	§A.2.3
<i>Low</i>	Important-instruments-only CMPI regression	§A.4.1
<i>Low</i>	Stock-Watson full mathematical derivation (Equations 1–3)	§A.1.1
<i>Low</i>	Ferrière suggestion: simple fiscal-monetary interaction exercise	§5 / future work
<i>Low</i>	Confirm Wind IRS data availability for full sample	§4.3

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